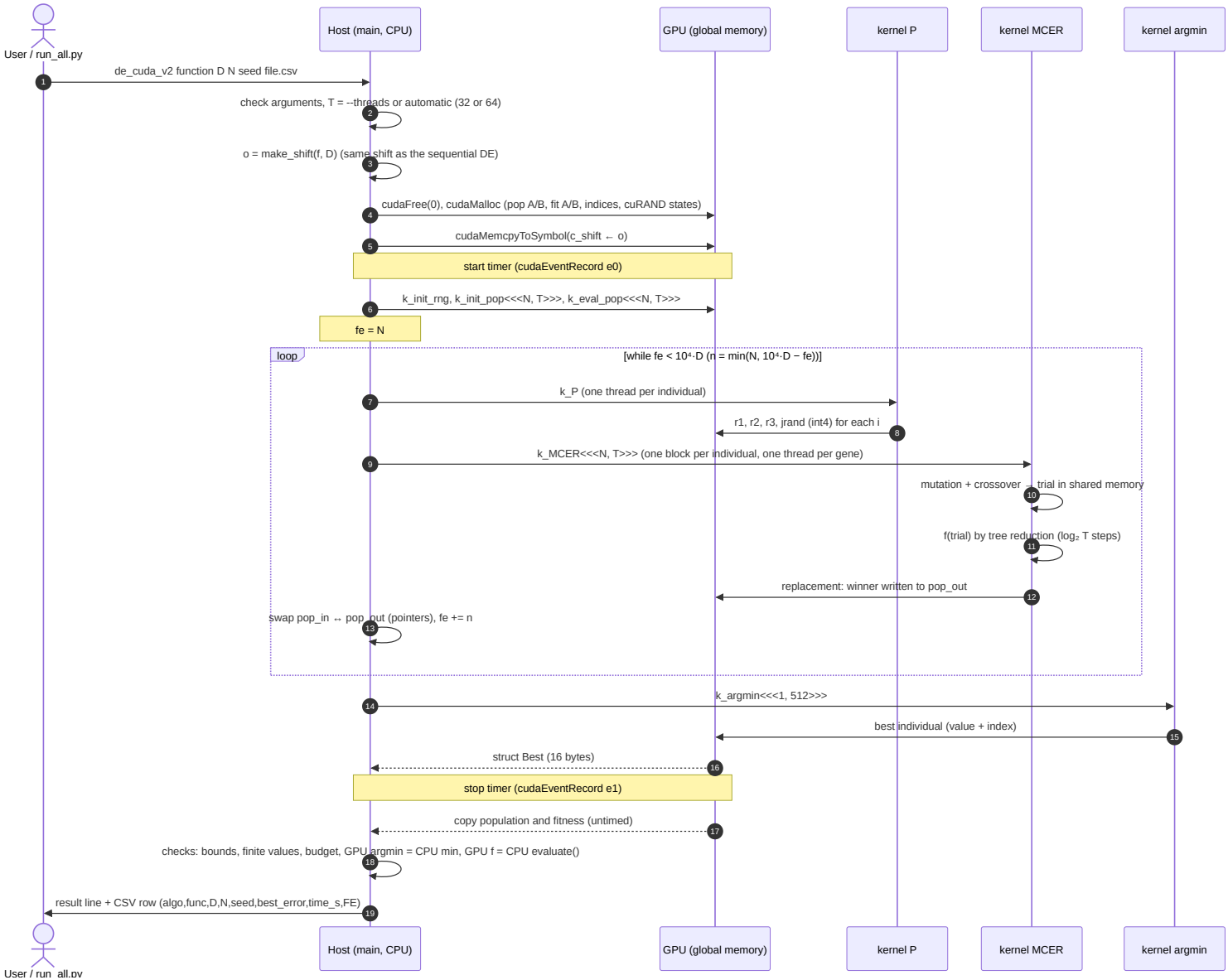
User / run_all.py
Host (main, CPU)
GPU (global memory)
kernel P
kernel MCER
kernel argmin
de_cuda_v2 function D N seed file.csv
check arguments, T = --threads or automatic (32 or 64)
o = make_shift(f, D) (same shift as the sequential DE)
cudaFree(0), cudaMalloc (pop A/B, fit A/B, indices, cuRAND states)
cudaMemcpyToSymbol(c_shift ← o)
start timer (cudaEventRecord e0)
k_init_rng, k_init_pop<<<N, T>>>, k_eval_pop<<<N, T>>>
fe = N
loop
[while fe < 10⁴·D (n = min(N, 10⁴·D − fe))]
k_P (one thread per individual)
r1, r2, r3, jrand (int4) for each i
k_MCER<<<N, T>>> (one block per individual, one thread per gene)
mutation + crossover
trial in shared memory
f(trial) by tree reduction (log₂ T steps)
replacement: winner written to pop_out
swap pop_in ↔ pop_out (pointers), fe += n
k_argmin<<<1, 512>>>
best individual (value + index)
struct Best (16 bytes)
stop timer (cudaEventRecord e1)
copy population and fitness (untimed)
checks: bounds, finite values, budget, GPU argmin = CPU min, GPU f = CPU evaluate()
result line + CSV row (algo,func,D,N,seed,best_error,time_s,FE)
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